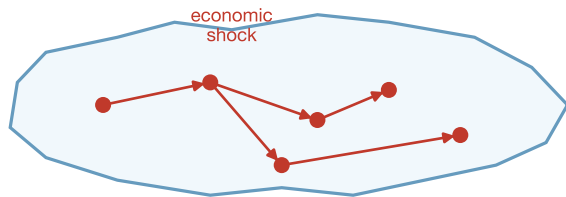


Polarity-Routed Cascade Diffusion for Crop Waste Prediction

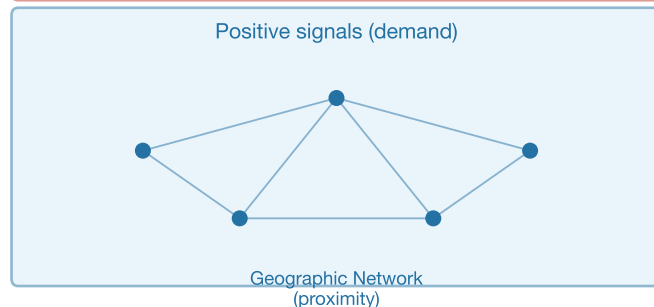
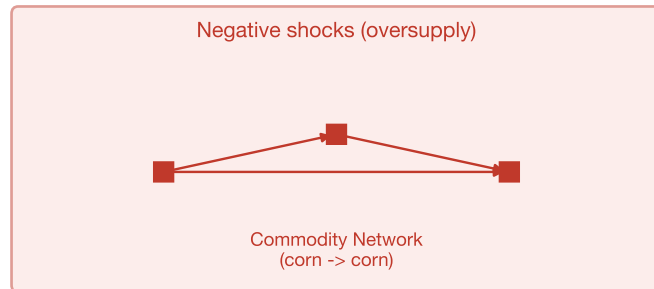
The Problem



\$19B in annual crop waste

Counties aren't independent —
shocks cascade across networks

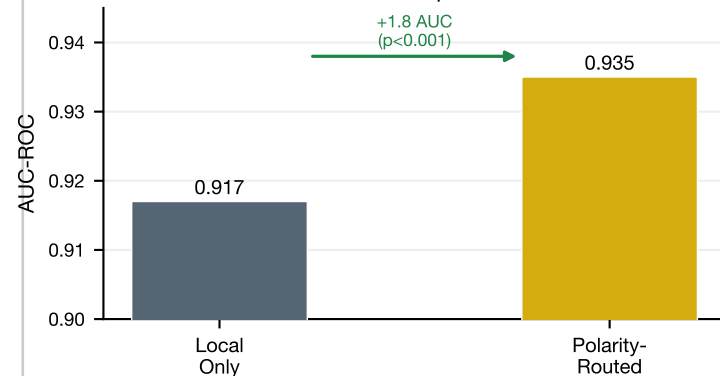
Our Insight



Different polarities -> Different topologies

Results

Model Comparison



638,622 USDA records

2,130 counties • 3 commodities

Temporal out-of-sample validation